Prof.Ajay Pashankar

Department Of Computer Science & Information Technology



ajaypashankar7

Prof.Ajay Pashankar

www.profajaypashankar.com

TY B.Sc. IT – Indian Knowledge Systems
Consolidated Question Bank – Module I & Module II

1. Explain the relevance of Artificial Intelligence concepts in the context of Ancient Indian Thought.
2. What is meant by a cognitive model? Explain its significance with suitable examples from Indian philosophical traditions.
3. Discuss the relationship between knowledge, perception, cognition, and intelligence in Indian philosophical systems.
4. Compare the concept of human intelligence in Ancient Indian Thought with modern Artificial Intelligence.
5. Explain the role of Nyāya logic and inference as an early framework for intelligent reasoning.
6. Differentiate between human cognition and machine intelligence with suitable examples.
7. Explain how ancient Indian theories of knowledge can contribute to the development of cognitive computing.
8. Application-based: If an AI system has to determine whether a statement is true using evidence and inference, explain which principles of Nyāya reasoning could be incorporated into its design.
9. Explain the concept of learning according to Ancient Indian Knowledge Systems.
10. Discuss the different stages involved in the acquisition of knowledge in traditional Indian learning systems.
11. Explain the relationship between memory, perception, attention, and learning.
12. Compare traditional Indian learning models with modern machine-learning models.
13. Explain the importance of the Guru–Shishya Parampara in knowledge transmission.
14. Discuss the role of observation, experience, repetition, and reasoning in the learning process.
15. Differentiate between human learning and machine learning.
16. Application-based: Design a conceptual AI learning model inspired by the traditional Indian approach to knowledge acquisition.
17. Discuss how cognition, learning, memory, perception, attention, and reasoning can collectively be represented in an intelligent computational system.
18. Critically discuss the similarities and differences between Ancient Indian theories of cognition and modern Artificial Intelligence.
19. Explain Nyāya reasoning and discuss its possible relevance to AI-based reasoning systems.
20. "Ancient Indian Knowledge Systems provide conceptual foundations, rather than direct equivalents, for several modern computational ideas." Discuss this statement with suitable examples.
21. Compare Nyāya inference and traditional Indian learning processes from the perspective of computational thinking.
22. Explain how a cognitive AI system could combine perception, memory, learning, reasoning, and decision-making.
23. Discuss the limitations of directly equating concepts from Ancient Indian philosophical traditions with modern AI and machine learning.
24. Explain the significance of knowledge acquisition in Ancient Indian Knowledge Systems and relate it to intelligent behaviour.
25. Discuss the role of perception (pratyakṣa) in knowledge acquisition and its relevance to computational cognition.
26. Explain how inference (anumāna) supports reasoning and decision-making in Indian philosophical thought.
27. Differentiate perception, inference, memory, and reasoning as cognitive processes.
28. Discuss how attention influences learning and knowledge acquisition in a cognitive model.
29. Explain the role of memory in learning and intelligent behaviour with suitable examples.
30. Discuss the role of repeated practice and experience in the formation of knowledge.
31. Explain how observation and experience contribute to learning in traditional Indian systems.
32. Compare rule-based reasoning in Nyāya with rule-based approaches used in Artificial Intelligence.

33. Discuss how evidence-based reasoning can be used to improve the reliability of an AI system.
34. Explain how human cognitive processes can inspire the design of intelligent agents.
35. Design a conceptual knowledge-acquisition pipeline based on perception, experience, memory, and reasoning.
36. Explain the role of reasoning and decision-making in both human intelligence and AI systems.
37. Discuss the similarities and differences between traditional knowledge transmission and machine-based knowledge acquisition.
38. Explain how Guru–Shishya Parampara supports structured, contextual, and experiential learning.
39. Discuss the importance of teacher guidance, practice, feedback, and experience in the learning process.
40. Explain how a modern AI system could model stages of knowledge acquisition inspired by Indian learning traditions.